249 Likes

Author **Topic: Starship specs - weight,**

fael097
Full Member

Posts: 1279
Rafael

Liked: 3619
Likes Given: 1028

Starship specs - weight, volumes, etc
« **on:** 02/04/2020 12:47 am »

Based on that coil tag and my own measurements of Mk1, I was trying to estimate some specs for SN1, mostly weight. I guess I went a bit overboard.

I have outlined a pretty accurate profile of the bulkheads and the fairing on cad, and got surface areas and volumes from it.

Thought this could be interesting for some and maybe get some peer review on the data.

I used the same steel thickness for everything, rings, bulkheads, fairing. Not sure right now if they're using thinner steel on top. Better data is very welcome.

Coils
Thickness: 3.97mm
Height: 1829mm
Length: 193m *(1 coil can make 6.83 rings)*
Weight: 11071kg
Volume: 1.4m^3
Density: 7907kg/m^3

Rings
Diameter: 9m
Circumference: 28.27m
Internal area *(Cross section)*: 63.6m^2
Surface area: 51.71m^2
Volume: 0.205m^3
Weight: 1621kg

Bulkheads

Surface area: 104.82m²
Volume: 0.419m³
Weight: 3313kg

Fairing *(curved nose section only)*
Surface area: 287.13m²
Volume: 1.135m³
Weight: 8974kg

LOX Header tank
Diameter: 3.14m
Internal volume: 18.67m³

Surface area *(without cone cap)*: 40.7m²
Material volume *(without cone cap)*: 0.08m³
Weight *(without cone cap)*: 632.6kg

CH4 Header tank
Diameter: 3.14m
Internal volume: 16.21m³

Weight *(Empty shell)*
3 Bulkheads: 9939kg
19 full rings: 30799kg
1.42m tall bottom ring *(77.64%)*: 1258.5kg
LOX Header tank: 632.6kg
Fairing: 8974kg
3 SL Raptors: 4500kg (estimated)
3 Vac Raptors: 6360kg (estimated)

Total weight: 56.1t *(Shell, tanks and engines* **-** *dry mass)*

Missing weight
-Welds
-Fins
-Actuators
-Batteries
-Plumbing
-Wiring
-Avionics
-Stringers
-TPS - **10t**
-COPVs
-Thrusters
-Landing legs
-[...]

Fuel/oxidizer densities
Densified LCH4: 451kg/m³
Densified LOX: 1230kg/m³

(source: http://spaceflight101.com/spx/its-booster)

Tank volumes & propellant mass
New SN8+ tanks
LOX header tank vol: 18.6m³ | mass:
LOX main 796.3m³
LOX Downcomer 2.8m³
CH4 header tank: 16.2m³
CH4 main 603.3m³
CH4 Downcomer 1.4m³

[old Mk1]
Mk1 tanks
Mk1 top tank (CH4): 600.08m³
Mk1 bottom tank (LOX): 808.88m³
Old capsule shaped header tanks: 15.7m³

Propellant mass
CH4 main tank: 263.8t
LOX main tank: 981.3t
Mass ratio: 1:3.71

LOX header tank: 22964.1kg
CH4 header tank: 7310.71kg

Thermal protection shield area
Half fuselage surface area: 655m²
Bottom fin single face area: 42.11m²
Top fin single face area: 17.75m²
Total TPS area: 774.72m²
Total TPS volume: 23.23m³
TPS density: **13kg/m²** *(may be outdated)*
Total TPS weight: **10t**

TPS hexagon tile dimension
Diameter: 35cm
Thickness: 3cm
Volume: 2387cm³ *(0.002387m³)*

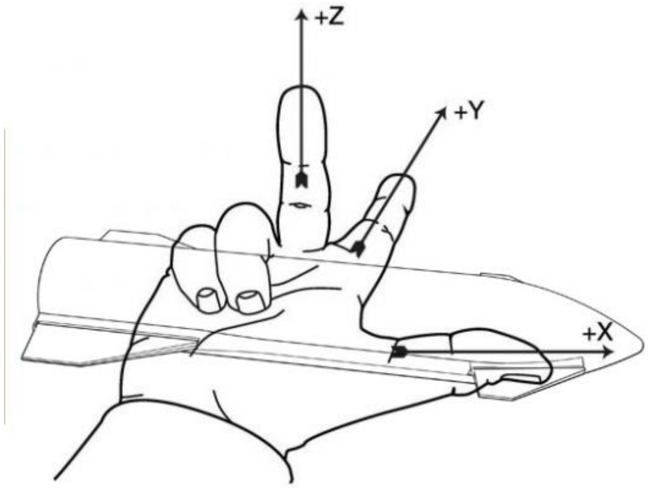
Starship local coordinate system
+X : Forward
-X : Aft
+Y : Left/Port
-Y : Right/Starboard
+Z : Up/Leeward
-Z : Down/Windward

****

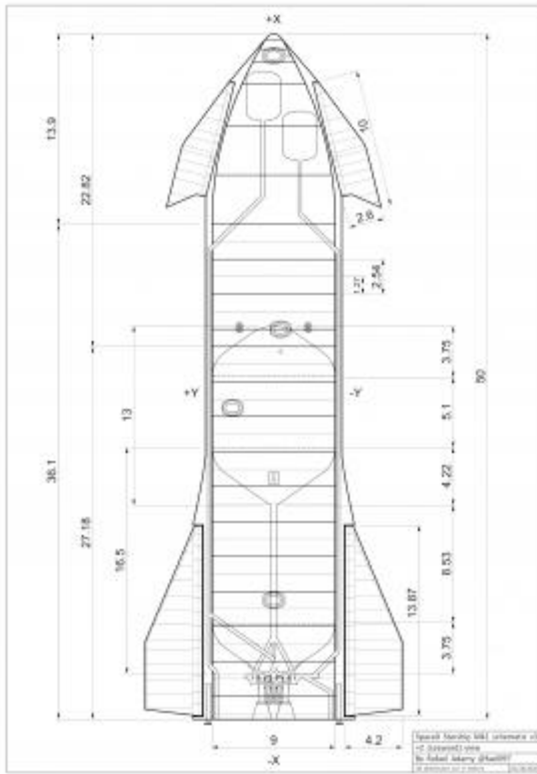
Attached images:
-Tag on the coil, pic by Mary - bocachicagal
-Mk1 leeward and aft diagrams

-SN1 diagram (subject to change)

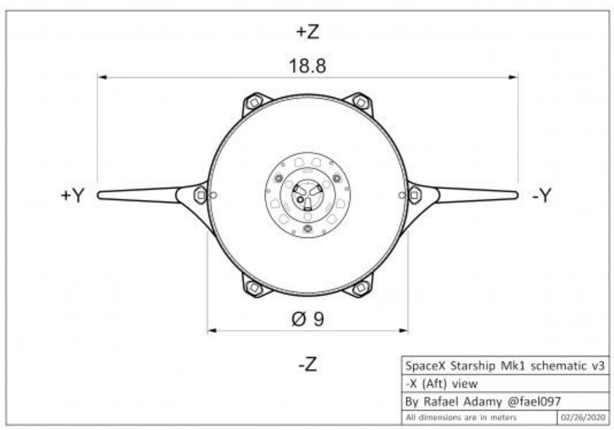
steel coil tag.jpg (195.6 kB, 884x664 - viewed 1124 times.)



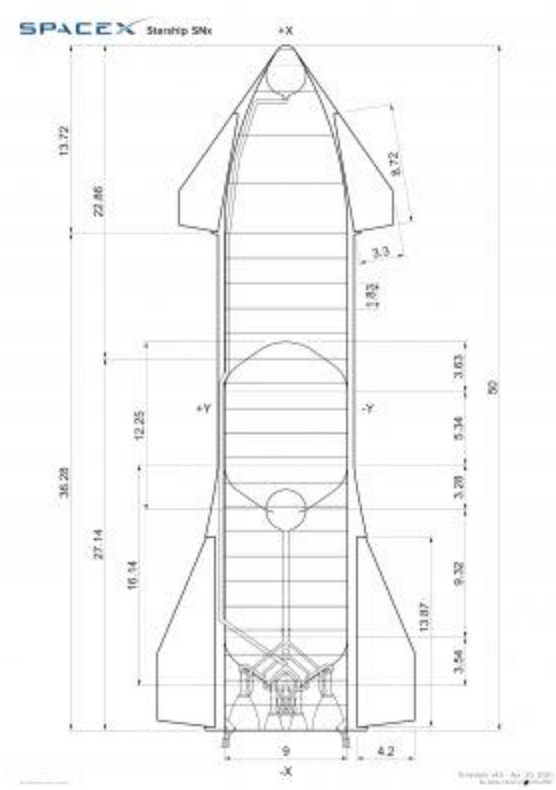
Right-Hand-Rule_starship.jpg (43.89 kB, 545x409 - viewed 1067 times.)



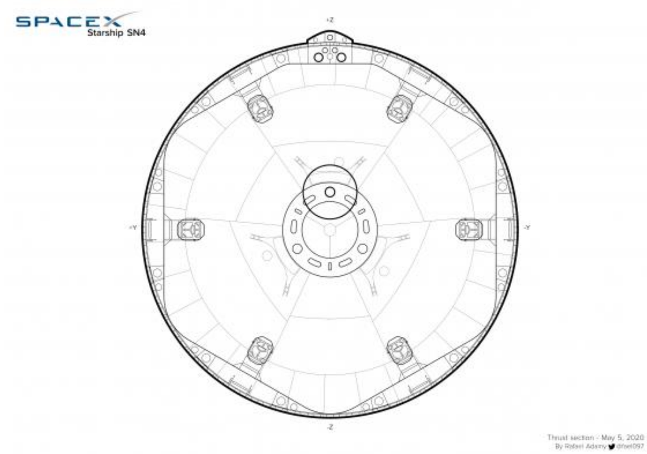
Mk1+Z_v3_fael097.png (637.82 kB, 3508x4961 - viewed 4554 times.)



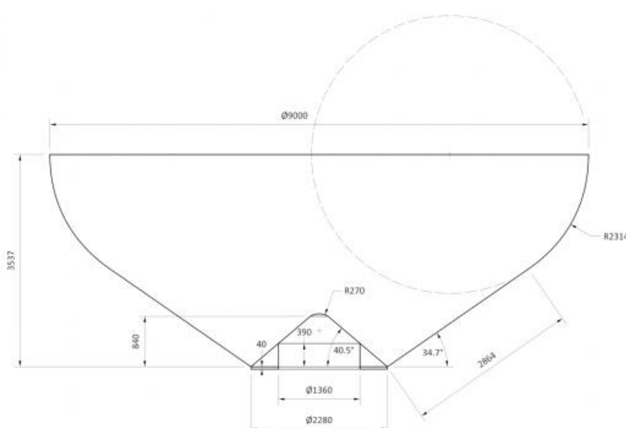
Mk1-X_v3_fael097.png (200.5 kB, 2480x1748 - viewed 1973 times.)



Starship_SNx+Z_v4_fael097.png (783.84 kB, 3508x4961 - viewed 7720 times.)



SN4_ThrustSection_fael097.png (1486.42 kB, 4961x3508 - viewed 1563 times.)



AftBulkDim.png (119.24 kB, 1920x1440 - viewed 1839 times.)

« *Last Edit: 11/05/2020 03:26 am by fael097* » Logged

hoardsbane, ThreadSnake, sebk and 28 others like this

Rafael Adamy

Nomadd

Senior Member

Posts: 9104
Virginia
Liked: 61610
Likes Given: 1411

Re: Starship specs - weight, volumes, etc

« **Reply #1 on:** 02/04/2020 01:33 am »

There are also 3mm and 2mm coils laying around.

Logged

Coastal Ron, livingjw, fael097 and 3 others like this

Those who danced were thought to be quite insane by those who couldn't hear the music.

Smrg

Member

Posts: 15
Toronto Canada
Liked: 8
Likes Given: 114

Re: Starship specs - weight, volumes, etc

« **Reply #2 on:** 02/04/2020 02:18 am »

Quote from: fael097 on 02/04/2020 12:47 am

> Bulkheads
> *********
> Surface area: 211.2m²
> Volume: 0.419m³
> Weight: 3313kg
>
> ...
>
> Weight (Empty shell)
> ********************
> 3 Bulkheads: 9939kg
> 19 full rings: 30799kg
> 1.42m tall bottom ring (77.64%): 1258.5kg
> LOX Header tank: 632.6kg
> Fairing: 8974kg
>
> ***
> *Total weight: 51.6t (Shell and tanks - empty)*
> ***

I think you tripled the bulkheads twice. They should be closer to 70 square meters, not 211 square meters each,
so we are down by 6.6 tons to 45 tons.

Logged

fael097 and ThreadSnake like this

geza

Full Member

Posts: 754
Budapest

Liked: 492
Likes Given: 90

Re: Starship specs - weight, volumes, etc

« **Reply #3 on:** 02/04/2020 08:05 am »

> **Quote from: Smrg on 02/04/2020 02:18 am**
>
> > **Quote from: fael097 on 02/04/2020 12:47 am**
> >
> > Bulkheads
> > *********
> > Surface area: 211.2m^2
> > Volume: 0.419m^3
> > Weight: 3313kg
> >
> > ...
> >
> > Weight (Empty shell)
> > *******************
> > 3 Bulkheads: 9939kg
> > 19 full rings: 30799kg
> > 1.42m tall bottom ring (77.64%): 1258.5kg
> > LOX Header tank: 632.6kg
> > Fairing: 8974kg
> >
> > ***
> > *Total weight: 51.6t (Shell and tanks - empty)*
> > ***
>
> I think you tripled the bulkheads twice. They should be closer to 70 square meters, not 211 square meters each, so we are down by 6.6 tons to 45 tons.

Cross section of the cylinder of 9 m diameter is 63.6 m2. If the bulkhead were a half sphere, its surface would be twice the cross section, i.e. 127.2 m2. The real surface area of the bulkhead should be in between these two values.

Logged

SteveU

Full Member

Posts: 395
New England
Liked: 443
Likes Given: 2685

Re: Starship specs - weight, volumes, etc

« **Reply #4 on:** 02/04/2020 10:55 am »

> **Quote from: geza on 02/04/2020 08:05 am**
>
> > **Quote from: Smrg on 02/04/2020 02:18 am**
> >
> > > **Quote from: fael097 on 02/04/2020 12:47 am**
> > >
> > > Bulkheads
> > > *********
> > > Surface area: 211.2m^2
> > > Volume: 0.419m^3
> > > Weight: 3313kg
> > >
> > > ...

Weight (Empty shell)
****
3 Bulkheads: 9939kg
19 full rings: 30799kg
1.42m tall bottom ring (77.64%): 1258.5kg
LOX Header tank: 632.6kg
Fairing: 8974kg

****
Total weight: 51.6t (Shell and tanks - empty)
****

I think you tripled the bulkheads twice. They should be closer to 70 square meters, not 211 square meters each,
so we are down by 6.6 tons to 45 tons.

Cross section of the cylinder of 9 m diameter is 63.6 m2. If the bulkhead were a half sphere, its surface would be twice the cross section, i.e. 127.2 m2. The real surface area of the bulkhead should be in between these two values.

Actually - only the top bulkhead is was close to being to spherical. The other two were more conical in shape.

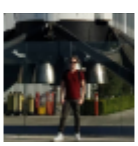
Also need to remember that the domes are significantly thicker than the normal sheeting. Also remember every weld seam has a stiffener added. I think Rafael's mass is probably a good estimate at this point.

[edit to add stiffener]

« *Last Edit: 02/04/2020 10:56 am by SteveU* » Logged

"Better a diamond with a flaw than a pebble without." - Confucius

fael097
Full Member



Posts: 1279
Rafael

Liked: 3619
Likes Given: 1028

Re: Starship specs - weight, volumes, etc
« **Reply #5 on:** 02/04/2020 11:35 am »

Quote from: Smrg on 02/04/2020 02:18 am

Quote from: fael097 on 02/04/2020 12:47 am

Bulkheads
****
Surface area: 211.2m²
Volume: 0.419m³
Weight: 3313kg

...

> > Weight (Empty shell)
> > *******************
> > 3 Bulkheads: 9939kg
> > 19 full rings: 30799kg
> > 1.42m tall bottom ring (77.64%): 1258.5kg
> > LOX Header tank: 632.6kg
> > Fairing: 8974kg
> >
> > ***
> > *Total weight: 51.6t (Shell and tanks - empty)*
> > ***
>
> I think you tripled the bulkheads twice. They should be closer to 70 square meters, not 211 square meters each, so we are down by 6.6 tons to 45 tons.

You're absolutely right, surface area is doubled. It calculated outer surface and inner surface as well. should be 104.82m². Same is true for fairing surface area. I corrected those on the original post.

However material volume is correct and so is weight (considering 3.97mm sheets)

Quote from: Nomadd on 02/04/2020 01:33 am

> There are also 3mm and 2mm coils laying around.

I remember seeing pics of those coils, but have we seen any evidence of those being used, and for what parts? Also do we know the thickness of those curved nose cone parts that were shipped to the site?

« *Last Edit: 02/04/2020 11:39 am by fael097* » Logged

Rafael Adamy

wannamoonbase
Elite Veteran
Senior Member
Posts: 6003
Denver, CO
Liked: 3711
Likes Given: 4810

Re: Starship specs - weight, volumes, etc
« **Reply #6 on:** 02/04/2020 12:16 pm »

Interesting exercise.

Is there a estimate for the heat shield? Surface area covered, multiple by thickness and density.

The bottom thrust structure and legs would seem to be a hard estimate, but that's going to have a lot of weight in it.

Looks like 100T could be possible.

Logged

fael097 likes this

I'm here for the mass driver.

fael097

Full Member

Posts: 1279

Rafael

Liked: 3619
Likes Given: 1028

Re: Starship specs - weight, volumes, etc

« **Reply #7 on:** 02/04/2020 02:05 pm »

> **Quote from: wannamoonbase on 02/04/2020 12:16 pm**
>
> Interesting exercise.
>
> Is there a estimate for the heat shield? Surface area covered, multiple by thickness and density.
>
> The bottom thrust structure and legs would seem to be a hard estimate, but that's going to have a lot of weight in it.
>
> Looks like 100T could be possible.

I could definitely estimate the surface area it will cover, and maybe thickness based on pictures, but I have no clue what the material density is

Logged

Rafael Adamy

ThatOldJanxSpirit

Full Member

Posts: 1115
Liked: 1729
Likes Given: 4581

Re: Starship specs - weight, volumes, etc

« **Reply #8 on:** 02/04/2020 05:13 pm »

> **Quote from: fael097 on 02/04/2020 02:05 pm**
>
> > **Quote from: wannamoonbase on 02/04/2020 12:16 pm**
> >
> > Interesting exercise.
> >
> > Is there a estimate for the heat shield? Surface area covered, multiple by thickness and density.
> >
> > The bottom thrust structure and legs would seem to be a hard estimate, but that's going to have a lot of weight in it.
> >
> > Looks like 100T could be possible.
>
> I could definitely estimate the surface area it will cover, and maybe thickness based on pictures, but I have no clue what the material density is

Best evidence is that the tiles are alumina / silica. The only dencity I have for the scintered material is 2.55 g/cc.

Awesome new thread Rafael.

Logged

fael097 and EeeVee3 like this

meekGee
Senior Member
Posts: 18042
N. California
Liked: 18305
Likes Given: 1504

Re: Starship specs - weight, volumes, etc
« **Reply #9 on:** 02/05/2020 12:27 pm »

> **Quote from: Nomadd on 02/04/2020 05:36 pm**
>
> > **Quote from: fael097 on 02/04/2020 11:35 am**
> >
> > > **Quote from: Nomadd on 02/04/2020 01:33 am**
> > >
> > > There are also 3mm and 2mm coils laying around.
> >
> > I remember seeing pics of those coils, but have we seen any evidence of those being used, and for what parts? Also do we know the thickness of those curved nose cone parts that were shipped to the site?
>
> Somebody thought that thickness might be mostly based on pressure and not vertical loads, so the 4mm could go all the way to the top of the tank section.
> Or, 3mm might be more than enough to hold pressure. Or a dozen other things. I'm not really sure how much more accurate my guesses would be than a monkey with a dartboard. Probably a little less.

I think the total pressure at any point is the head pressure (under the momentary g force) plus the ullage pressure. This is acting in all directions, and will be highest at the bottom, right?

Plus the upper tank is pushing down, through the walls. Some of that compressive stress is actually cancelled out by the tension induced by the pressure.

Also propellant density changes, and possibly ullage pressure too.

So IMO it might be that due to optimization and weld strength the entire bottom tank is one constant thickness, but then it can start tapering down as you go up the rocket.

The payload bay itself, that's a different story. No pressure stabilization at launch time, lots of internal structure, hard mount points for heavy stuff.. who knows.

Logged

ABCD - Always Be Counting Down

ThatOldJanxSpirit

Full Member

Posts: 1115
Liked: 1729
Likes Given: 4581

Re: Starship specs - weight, volumes, etc
« **Reply #10 on:** 02/05/2020 01:52 pm »

> **Quote from: ThatOldJanxSpirit on 02/04/2020 05:13 pm**
>
> > **Quote from: fael097 on 02/04/2020 02:05 pm**
> >
> > > **Quote from: wannamoonbase on 02/04/2020 12:16 pm**
> > >
> > > Interesting exercise.
> > >
> > > Is there a estimate for the heat shield?
> > > Surface area covered, multiple by thickness and density.
> > >
> > > The bottom thrust structure and legs would seem to be a hard estimate, but that's going to have a lot of weight in it.
> > >
> > > Looks like 100T could be possible.
> >
> > I could definitely estimate the surface area it will cover, and maybe thickness based on pictures, but I have no clue what the material density is
>
> Best evidence is that the tiles are alumina / silica. The only dencity I have for the scintered material is 2.55 g/cc.
>
> Awesome new thread Rafael.

A bit more digging on the ceramics. Bulk density is around 2.5-3 g/cc but this doesn’t take account of voidage. At the extreme these materials can be blown to 95% voidage. Looking at commercial products Zircar Ceramics do a wide range with good strength, sustained max temperatures up to 1450 C and density down in the 0.3 - 0.5 g/cc range. This is probably an appropriate approximation.

Did anybody estimate how thick the test tiles were?

Logged

livingjw

Senior Member

Posts: 2382
New World

Re: Starship specs - weight, volumes, etc
« **Reply #11 on:** 02/05/2020 02:43 pm »

- TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.

- What do you calculate for the fuselage and fin areas?

Liked: 5911
Likes Given: 2928

John

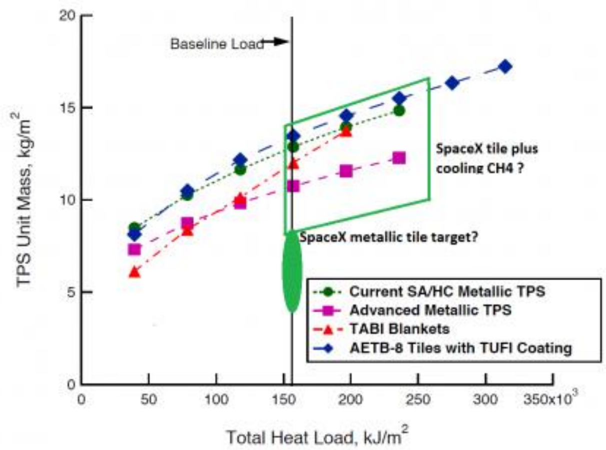


Figure 21. TPS unit mass as a function of total heat load using scaled heat loads from the Access to Space vehicle.

TPS_Wt_Trends.png (96.46 kB, 1157x1036 - viewed 990 times.)

« *Last Edit: 02/05/2020 02:51 pm by livingjw* » Logged

PADave, fael097 and volker2020 like this

wannamoonbase

Elite Veteran
Senior Member
Posts: 6003
Denver, CO
Liked: 3711
Likes Given: 4810

Re: Starship specs - weight, volumes, etc
« **Reply #12 on:** 02/05/2020 03:22 pm »

Quote from: livingjw on 02/05/2020 02:43 pm

> - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
>
> - What do you calculate for the fuselage and fin areas?
>
> John

John,

Thanks that's a lot lower than I expected.

Doing rough math on covering half of a 9 meter diameter vehicle 118 meters long with 13 kg/m^2 tile = 21,700 kg.

Logged

I'm here for the mass driver.

PADave

Member
Posts: 40
York PA
Liked: 44

Re: Starship specs - weight, volumes, etc
« **Reply #13 on:** 02/05/2020 03:25 pm »

Quote from: livingjw on 02/05/2020 02:43 pm

Likes Given: 313

> - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
>
> - What do you calculate for the fuselage and fin areas?
>
> John

Don't you think it will run heaver than that since they are shooting for non-ablative tiles. I guess I really don't know but I would have suspected that making them more durable would incur a mass penalty... or would it be offset by the higher temperature that they can tolerate?

Logged

rakaydos

Senior Member

Posts: 2843
Liked: 1876
Likes Given: 70

Re: Starship specs - weight, volumes, etc

« **Reply #14 on:** 02/05/2020 03:36 pm »

Quote from: livingjw on 02/05/2020 02:43 pm

> - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
>
> - What do you calculate for the fuselage and fin areas?
>
> John

That graphic looks out of date- it seems to be comparing to "metallic TPS" and "Metallic + CH4 cooling", when we have been told that SpaceX has selected a non-ablative ceramic TPS to be bolted over the steel hull. (I have seen speculation to the effect of a TUFROC based tile, but without as many insulating layers because the steel hull can handle a higher temperature than an aluminum airframe.)

Logged

envy887

Senior Member

Posts: 8551
Liked: 7358
Likes Given: 3027

Re: Starship specs - weight, volumes, etc

« **Reply #15 on:** 02/05/2020 03:40 pm »

Quote from: wannamoonbase on 02/05/2020 03:22 pm

> **Quote from: livingjw on 02/05/2020 02:43 pm**
>
> > - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
> >
> > - What do you calculate for the fuselage and fin areas?
> >
> > John
>
> John,
>
> Thanks that's a lot lower than I expected.

> Doing rough math on covering half of a 9 meter diameter vehicle 118 meters long with 13 kg/m^2 tile = 21,700 kg.

118 m? Starship is only ~35 m long at 9 m diameter, plus a smaller average diameter ~20 m long cone section.

SuperHeavy will not have tile TPS.

Logged

fael097

Full Member

Posts: 1279

Rafael

Liked: 3619
Likes Given: 1028

Re: Starship specs - weight, volumes, etc
« **Reply #16 on:** 02/05/2020 03:41 pm »

Quote from: livingjw on 02/05/2020 02:43 pm

> - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
>
> - What do you calculate for the fuselage and fin areas?
>
> John

Half fuselage has a surface area of 655m^2. I'm not considering raceway covers, shouldn't matter that much.
Surface area of one side of a bottom fin is 42.11m^2 and top fin is 17.75m^2.

So total heat shield area should be around 774.72m^2

Quote from: ThatOldJanxSpirit on 02/05/2020 01:52 pm

> A bit more digging on the ceramics. Bulk density is around 2.5-3 g/cc but this doesn't take account of voidage. At the extreme these materials can be blown to 95% voidage. Looking at commercial products Zircar Ceramics do a wide range with good strength, sustained max temperatures up to 1450 C and density down in the 0.3 - 0.5 g/cc range. This is probably an appropriate approximation.
>
> Did anybody estimate how thick the test tiles were?

Heat shield tiles seem to be 35cm in diameter and 3cm thick, so volume of a tile should be 2387cm^3 or 0.002387m^3

Logged

Rafael Adamy

ThatOldJanxSpirit

Re: Starship specs - weight, volumes, etc
« **Reply #17 on:** 02/05/2020 04:21 pm »

Full Member

Posts: 1115
Liked: 1729
Likes Given: 4581

Quote from: fael097 on 02/05/2020 03:41 pm

> **Quote from: livingjw on 02/05/2020 02:43 pm**
>
> - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
>
> - What do you calculate for the fuselage and fin areas?
>
> John

Half fuselage has a surface area of 655m^2. I'm not considering raceway covers, shouldn't matter that much. Surface area of one side of a bottom fin is 42.11m^2 and top fin is 17.75m^2.

So total heat shield area should be around 774.72m^2

> **Quote from: ThatOldJanxSpirit on 02/05/2020 01:52 pm**
>
> A bit more digging on the ceramics. Bulk density is around 2.5-3 g/cc but this doesn't take account of voidage. At the extreme these materials can be blown to 95% voidage. Looking at commercial products Zircar Ceramics do a wide range with good strength, sustained max temperatures up to 1450 C and density down in the 0.3 - 0.5 g/cc range. This is probably an appropriate approximation.
>
> Did anybody estimate how thick the test tiles were?

Heat shield tiles seem to be 35cm in diameter and 3cm thick, so volume of a tile should be 2387cm^3 or 0.002387m^3

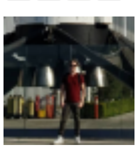
So around 10 tonne assuming 13 kg/m3 and 12 tonnes assuming 3cm of alumina silica tile. Consistent, and a lot less than I thought.

For info, I'm assuming alumina silica because there was a job posting for a manager of alumina silica tile production.

Logged

fael097

Full Member



Posts: 1279

Rafael

Re: Starship specs - weight, volumes, etc

« **Reply #18 on:** 02/05/2020 04:46 pm »

Quote from: ThatOldJanxSpirit on 02/05/2020 04:21 pm

> **Quote from: fael097 on 02/05/2020 03:41 pm**
>
> > **Quote from: livingjw on 02/05/2020 02:43 pm**
> >
> > - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including

Liked: 3619
Likes Given: 1028

> > > fins.
> > >
> > > - What do you calculate for the fuseloge and fin areas?
> > >
> > > John
> >
> > Half fuselage has a surface area of $655m^2$. I'm not considering raceway covers, shouldn't matter that much.
> > Surface area of one side of a bottom fin is $42.11m^2$ and top fin is $17.75m^2$.
> >
> > So total heat shield area should be around $774.72m^2$
> >
> > **Quote from: ThatOldJanxSpirit on 02/05/2020 01:52 pm**
> >
> > > A bit more digging on the ceramics. Bulk density is around 2.5-3 g/cc but this doesn't take account of voidage. At the extreme these materials can be blown to 95% voidage. Looking at commercial products Zircar Ceramics do a wide range with good strength, sustained max temperatures up to 1450 C and density down in the 0.3 - 0.5 g/cc range. This is probably an appropriate approximation.
> > >
> > > Did anybody estimate how thick the test tiles were?
> >
> > Heat shield tiles seem to be 35cm in diameter and 3cm thick, so volume of a tile should be $2387cm^3$ or $0.002387m^3$
>
> So around 10 tonne assuming 13 kg/m3 and 12 tonnes assuming 3cm of alumina silica tile. Consistent, and a lot less than I thought.
>
> For info, I'm assuming alumina silica because there was a job posting for a manager of alumina silica tile production.

Considering total TPS volume of $23.34m^3$, if you assumed $3g/cm^3$ ($3000kg/m^3$) total TPS weight would be 69.72t. So shell, bulkheads and TPS alone would weight 121.32t already, above the targeted 120t dry mass.

John's graph showed ~13kg/m² so that should include the necessary volume to given area. 10t seems right according to that graph, and it would result a total weight of 61.6t for shell, bulkheads and TPS. I'll consider that until we have better data on TPS density.

Logged

Rafael Adamy

ThatOldJanxSpirit

Full Member

Posts: 1115

Liked: 1729

Likes Given: 4581

Re: Starship specs - weight, volumes, etc

« **Reply #19 on:** 02/05/2020 04:57 pm »

> **Quote from: fael097 on 02/05/2020 04:46 pm**
>
> > **Quote from: ThatOldJanxSpirit on 02/05/2020 04:21 pm**
> >
> > > **Quote from: fael097 on 02/05/2020 03:41 pm**
> > >
> > > > **Quote from: livingjw on 02/05/2020 02:43 pm**
> > > >
> > > > - TPS weight trends. I would guess 10-13 kg/m^2 on average for windward side including fins.
> > > >
> > > > - What do you calculate for the fuselage and fin areas?
> > > >
> > > > John
> > >
> > > Half fuselage has a surface area of 655m². I'm not considering raceway covers, shouldn't matter that much.
> > > Surface area of one side of a bottom fin is 42.11m² and top fin is 17.75m².
> > >
> > > So total heat shield area should be around 774.72m²
> > >
> > > > **Quote from: ThatOldJanxSpirit on 02/05/2020 01:52 pm**
> > > >
> > > > A bit more digging on the ceramics. Bulk density is around 2.5-3 g/cc but this doesn't take account of voidage. At the extreme these materials can be blown to 95% voidage. Looking at commercial products Zircar Ceramics do a wide range with good strength, sustained max temperatures up to 1450 C and density down in the 0.3 - 0.5 g/cc range. This is probably an appropriate approximation.
> > > >
> > > > Did anybody estimate how thick the test tiles were?
> > >
> > > Heat shield tiles seem to be 35cm in diameter and 3cm thick, so volume of a tile should be 2387cm³ or 0.002387m³
> >
> > So around 10 tonne assuming 13 kg/m3 and 12 tonnes assuming 3cm of alumina silica tile. Consistent, and a lot less than I thought.
> >
> > For info, I'm assuming alumina silica because there

> > was a job posting for a manager of alumina silica tile production.
>
> Considering total TPS volume of 23.34m³, if you assumed 3g/cm³ (3000kg/m³) total TPS weight would be 69.72t. So shell, bulkheads and TPS alone would weight 121.32t already, above the targeted 120t dry mass.
>
> John's graph showed ~13kg/m² so that should include the necessary volume to given area. 10t seems right according to that graph, and it would result a total weight of 61.6t for shell, bulkheads and TPS. I'll consider that until we have better data on TPS density.

I was assuming 0.5g/cc which is the high end for COTS materials. 3g/cc is for bulk alumina silica material with no porosity.

Logged

0